

Circle the correct answer: (1 point / question)

- 1- Fragmentation is the process of breaking down a database into a set of sub-databases.
a- True.
b- False.
- 2- The rules of fragmentation are: completeness, union and reconstruction.
a- True
b- False
- 3- Transparency for the user and the autonomy of each site are two of the objectives of distributed SGBD.
a- True
b- False
- 4- The operator of the Horizontal fragmentation is the selection (σ).
a- True.
b- False.
- 5- The operator of the reconstruction of the Horizontal fragmentation is the Join ($\bowtie$).
a- True
b- False
- 6- The operator of the distribution of attributes (vertical fragmentation) is the projection (π)
a- True
b- False
- 7- The operator of the reconstruction of the distribution of attributes (vertical fragmentation) is the Semi-Join ($\ltimes$).
a- True
b- False
- 8- Distribution of values (Mixed fragmentation): It is the combination of the two previous fragmentation, horizontal and vertical.
a- True
b- False
- 9- The partitioning operation is a combination of projections and selections.
a- True
b- False
- 10- Let $R(a, b)$, the values of a and b are positive.
The expression $SF(\sigma_{(a=1) \wedge (b>1)}(R))$ is equivalent to:
 $SF(\sigma_{a=1}(R)) + SF(\sigma_{b>1}(R))$
a- True
b- False
- 11- Let $R(a, b)$, the values of a and b are positive.
The expression $SF(\sigma_{(a=1) \wedge (b>1)}(R))$ is equivalent to:
 $SF(\sigma_{a=1}(R)) * (1 - SF(\sigma_{b \leq 1}(R)))$
a- True
b- False
- 12- Prolog is a programming language:
a- An object oriented language.
b- A procedural language.
c- A logical programming language.
d- An http tag.
- 13- Which of the following RDBMS is qualified as object oriented?
a- INFORMIX
b- ORACLE
c- O2
- 14- What is a relational diagram?
a- A schema that logically represents the tables in a database.
b- A representation of a set of interrelated relationships.
c- A diagram which represents the flow of data between the actors.
d- A diagram that explains the interactions between users.

15- What is "serialization" ?

- a- Stacking and unstacking objects
- b- Saving objects in a file or for transport
- c- Assigning serial number to objects

16- What represents an attribute in a relational database?

- a- Table
- b- Line
- c- Column
- d- Object

17- Let R (a, b) S (a, c, d) and the query:

Select S.d

From R, S

Where R.a = S.a and R.b=1

We apply, on this query, the heuristic rule of restructuring which consists in treating first the most reducing operations. The resulting expression is:

- a- $\pi_d(\sigma_{(b=1)}(R \bowtie_a S))$
- b- $\pi_d[(\pi_a(\sigma_{(b=1)}(R)) \bowtie_a (\pi_{a,d}(S))]$
- c- $(\sigma_{(b=1)}(R)) \bowtie_a (\pi_d(S))$

18- ODL supports which of the following types of association relationships?

- a- Unary
- b- Unary and Binary
- c- Unary and Binary and Ternary
- d- Unary and Binary and Ternary and higher

19- An extent is which of the following?

- a- A keyword that indicates that the subclass inherits from a superclass
- b- A keyword that indicates that the superclass inherits from a subclass
- c- The set of all instances of a class within a database
- d- Only one instance of a class within a database

20- Using ODL, you can define which of the following?

- a- Attribute
- b- Structure
- c- Operation
- d- All of the above